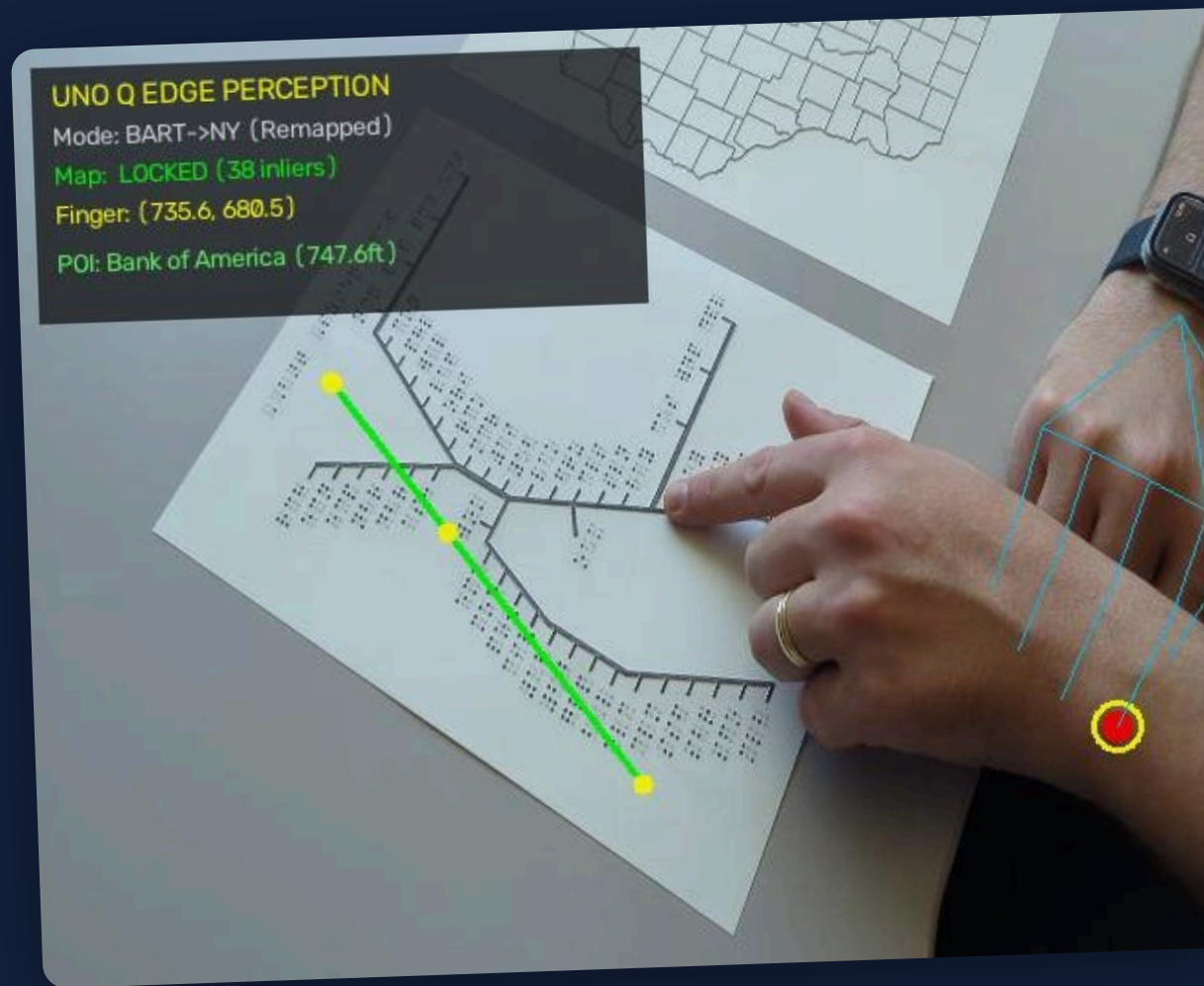


MAPIO · QUALCOMM EDITION

Explore a map with your finger. Hear what's there.

Audio-tactile maps with fingertip tracking, a local LLM on Snapdragon X Elite, and a sub-\$60 standalone client on Arduino UNO Q.



MOTIVATION

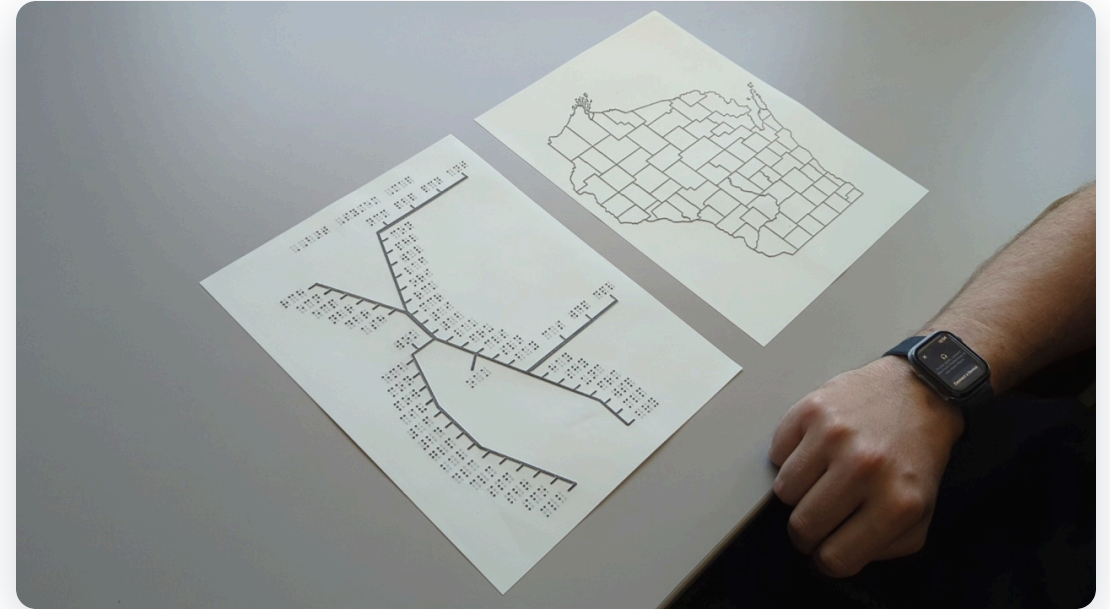
Great for navigation. Inaccessible for exploration.

Digital maps

Turn-by-turn works with a screen reader. But **exploring** — what's around here, how the streets lie, what's two blocks over — is a visual act. A blind user gets a list, never a layout.

Tactile maps

Embossed streets give real spatial layout under the fingers. The price is **information density**: a sheet fits a dozen braille labels, and nothing else — no store names, hours, distances, or "what is this?"

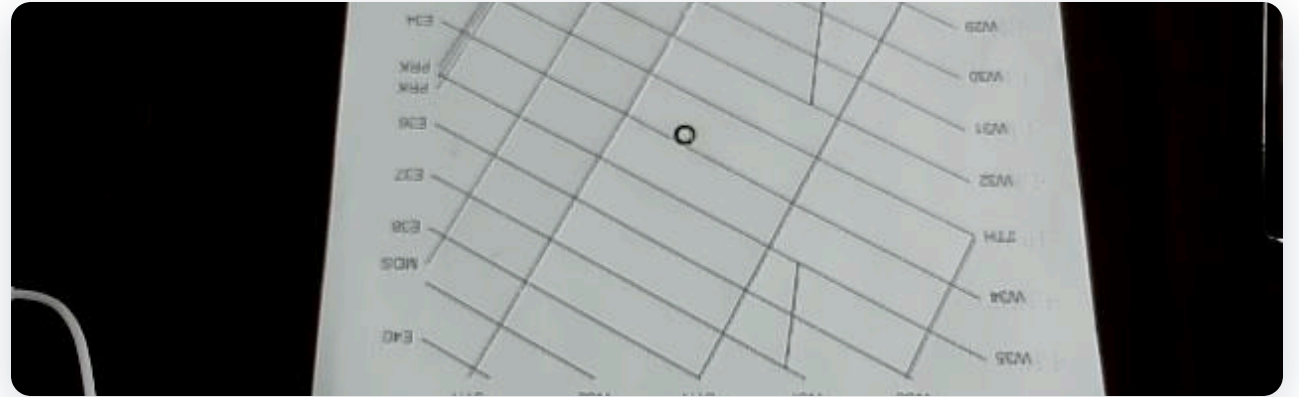


Embossed tactile maps: excellent layout, a handful of labels per sheet.

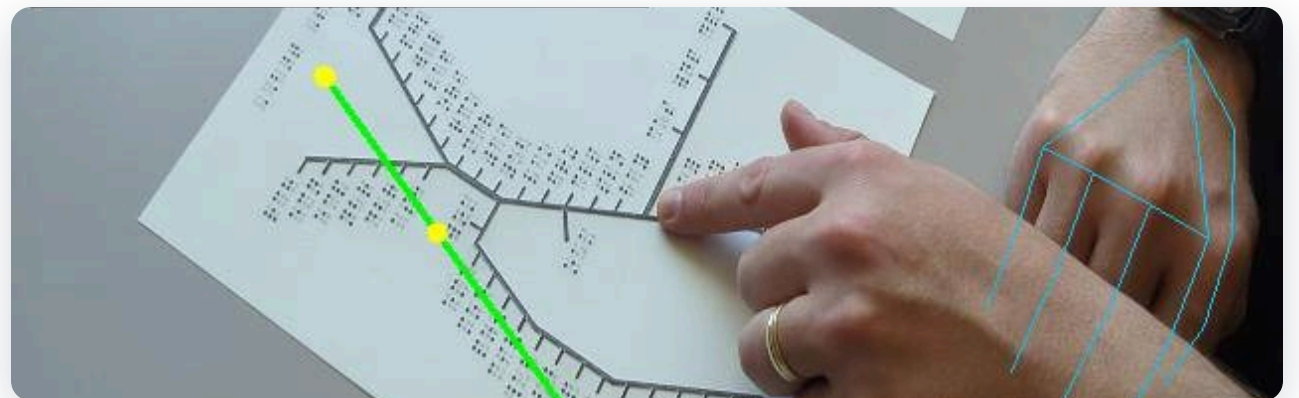
THE IDEA

Your fingertip is your avatar. The map's information is virtual — and audio.

- A camera looks down at an ordinary tactile map. **SIFT homography** locks the sheet; **MediaPipe** tracks the index fingertip.
- Fingertip → map coordinates → a topological **street & POI graph**. Wherever you touch, you hear what's there.
- The tactile sheet stays sparse and cheap. The **audio layer is unbounded**: names, distances, details, routes — density limited by curiosity, not emboss space.
- Nothing on the map changes. Any existing tactile map becomes an interactive one.



Camera view: a Lighthouse-SF NYC street map (20 W 34th St).



Live perception on the UNO Q: map locked, fingertip mapped, nearest POI resolved.

ASK ANYTHING

Open-ended questions from an LLM. Deterministic answers from tools.

"What's the closest place to get coffee, and how far is it from where my finger is?"

Free-form questions are what make exploration feel natural. But distances, directions and "am I here?" must be **right**, not plausible. So the model doesn't guess — it **calls tools on the map graph** and speaks the result.

get_distance

get_nearby_points_of_interest

am_i_at_point_of_interest

get_point_of_interest_details

guide_to_destination

Runs on the laptop's NPU, not in the cloud

Gemma 4 E2B (Q4) served by **Qualcomm AI Hub GenieX** on the Snapdragon X Elite — an OpenAI-compatible endpoint on localhost. Private, offline, no per-query cost.

Position is always in the prompt

Every question carries the fingertip's map coordinates, so "here", "this", and "nearby" resolve to real places. Tool results are read back as spoken announcements with priority and category, the same pipeline as touch feedback.

HARDWARE

Two Qualcomm chips, two roles.

Snapdragon X Elite — the brain

1

local LLM server per room

- GenieX serving Gemma 4 on Hexagon NPU / Adreno GPU
- Speech-to-text + question answering
- Never leaves the building

Arduino UNO Q — the station

<\$60

per map station (QRB2210, Linux)

- Runs the whole perception loop on-device: SIFT lock, MediaPipe hand tracking, graph lookup, TTS
- Wi-Fi + Bluetooth on board
- Camera + power = a standalone map

Ray-Ban Meta — the voice

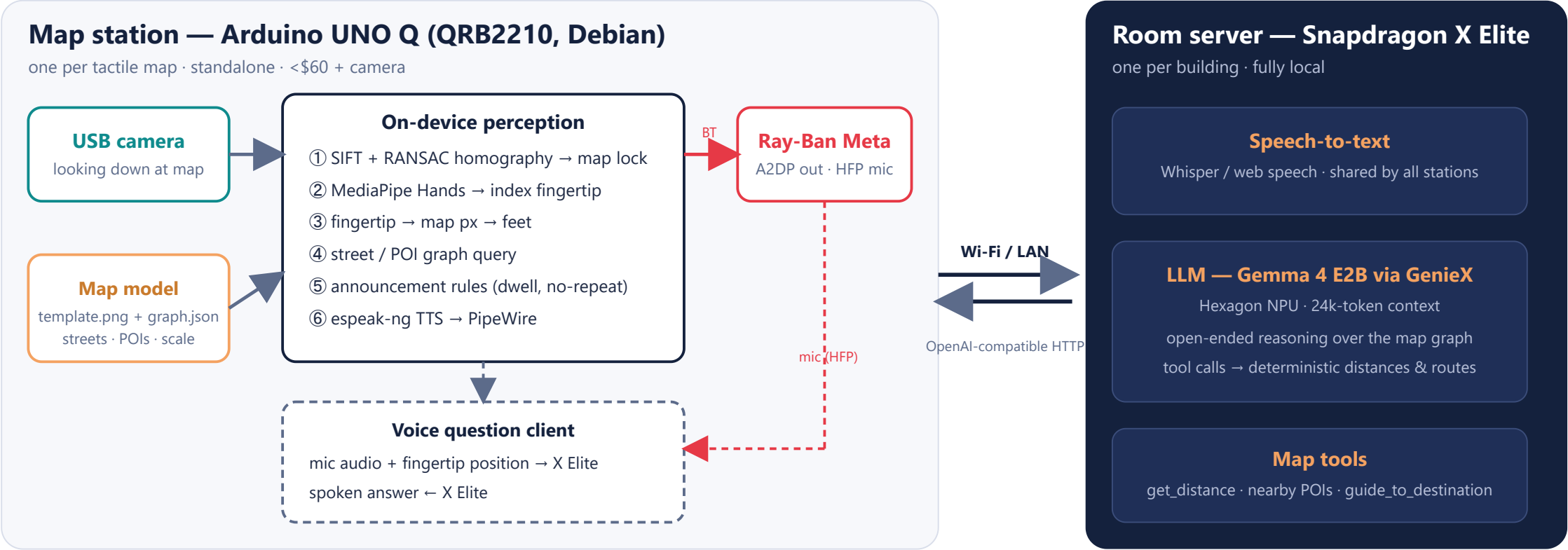
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wires to the user

- Paired directly to the UNO Q over Bluetooth A2DP (BlueZ + PipeWire)
- Open-ear audio: hear the map and the room
- Mic ready for voice questions

SYSTEM ARCHITECTURE

Perception at the edge, reasoning in the room.



DESIGN CHOICE

A laptop alone works. The UNO Q changes what's possible.

MapIO runs end-to-end on the X Elite (camera, LLM, speech) — that is a valid deployment. Splitting perception onto a UNO Q per map is what makes it deployable at scale.

	LAPTOP-ONLY	UNO Q STATION + X ELITE SERVER
Cost per map	a full laptop, dedicated	<\$60 board + webcam; one laptop shared
Footprint	laptop on the table, lid open	board tapes under the table; power bank runs it
Setup by staff	launch app, pick camera, keep awake	plug in → boots → speaks "Map detected"
Touch feedback latency	local	local — perception never crosses the network
Many maps, one room	N laptops, N LLM copies	N boards, 1 LLM — questions queue, touch never waits
Privacy	on-device	on-premises; nothing leaves the LAN
Audio to user	laptop speaker / wired	Bluetooth to the user's own glasses

FUTURE DIRECTIONS

Many stations. One brain. Then no brain needed.

1 · Fleet of stations

A museum floor or a classroom: every tactile map gets a UNO Q + camera. All of them speak to **one X Elite** for STT and Q&A. Per-map cost stays flat; the server cost is paid once.

2 · STT on the UNO Q itself

Whisper-tiny class models fit the QRB2210. With the glasses' mic over HFP, the station transcribes locally and only sends **text + position** to the server — lower bandwidth, lower load, works over weak Wi-Fi.

3 · Unified Q&A across maps

The server sees every station's position: "where is the exhibit I was at before?" across rooms, shared routes between maps, and a per-user session that follows the visitor from map to map.

Also on the list

- Boot-to-app image for the UNO Q (systemd unit, auto-pair to glasses)
- Powered USB-C hub → camera hosted by the board, no laptop in the loop
- Map authoring: photograph any tactile map + OSM extract → new model

Why this matters

Tactile maps exist in transit hubs, campuses and museums today, mostly unused because they say so little. A camera and a board per sheet turns each one into a place you can ask questions of — without touching the map itself.

DEMO

See it work.

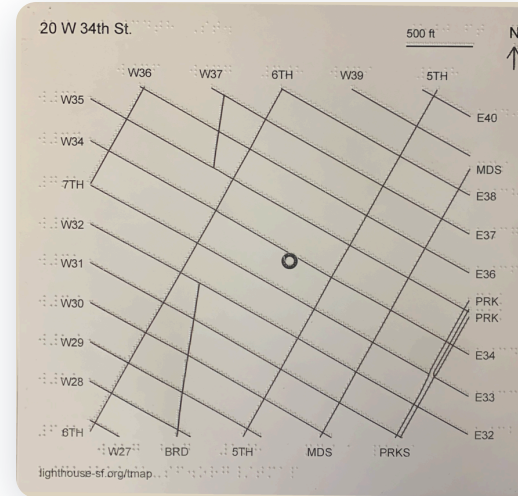
Demo video: NYC street map on a Lighthouse-SF tactile sheet, perception on the UNO Q, speech in the Ray-Ban Meta glasses, questions answered by Gemma on the X Elite.

Stack

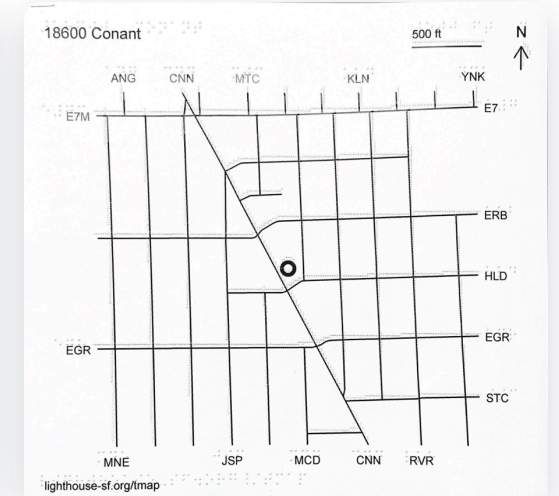
OpenCV SIFT · MediaPipe Hands · BlueZ 5.82 + PipeWire · espeak-ng · Qualcomm AI Hub GenieX · Gemma 4 E2B (Q4) · Python on Debian (UNO Q) and Windows on ARM64 (X Elite)

Maps ready today

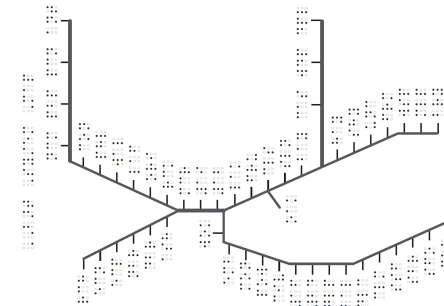
New York (Midtown), Detroit (Conant), BART system map



New York



Detroit · Conant



BART